# 1.4 Hazards, opportunities and management

## Opportunities of living near a river

Living near a river can offer various opportunities. These include:

- Water supply: rivers provide fresh water that can be used for drinking, **irrigation**, and industry.
- Energy: hydroelectric power (HEP) is produced when flowing water is used to turn turbines, generating electricity. Small and large HEP stations can provide a sustainable and renewable energy source.
- Transportation: rivers can support the movement of people and goods. In some parts of the world, people still rely on rivers for trade and transportation.
- Recreation: leisure activities such as fishing, boating, and swimming can improve peoples' mental and physical health. These activities can also provide an income for local people and contribute to a region's tourism industry.

**Source** A: Srinakarin Dam and HEP station, Thailand



**TASK 1: Study Source A**

**a** Explain how the water in the lake at the top is replenished.

**b** Suggest why HEP might be considered better than burning fossil fuels.

**c** Identify any potential disadvantages to HEP.

## Hazards of living near a river

Living near a river can also bring hazards. Rivers can burst their banks and flood the surrounding landscape, particularly during times of heavy rainfall or snowmelt. Continuous erosion by rivers can lead to the loss of valuable land, such as farmland, private properties, and **infrastructure**. Industrial discharge, runoff from farmland, and sewage can also contaminate rivers and pose health risks.

**Task 2**

**a** Can you think of any more opportunities and hazards associated with rivers?

**b** Identify which opportunities and hazards associated with rivers:

  **i** differ between different countries

  **ii** change over time

  Explain how.

# River flooding and pollution

River flooding has natural and human causes. Examples of natural causes of flooding include heavy rainfall and rapid snowmelt. Humans can also create floods or make them worse. Activities such as deforestation and **urbanisation** decrease the amount of rainfall that can infiltrate and percolate into the landscape, increasing run-off. Climate change is also leading to higher rates of snowmelt and increased rainfall in some places, both of which can cause flooding.

**Source B**: Flooding in Cumbria, UK



**TASK 3: Study Source B**

**a** Work with a partner. Describe the damage that can be seen in the image.

**b** List the 'costs' of – or damage from – flooding under the following headings:

- **i** Social
- **ii** Economic
- **iii** Environmental
- **iv** Peer-assess the answers of another pair in the class by comparing their answers with yours. Are there any areas where your answers differ and, if so, why?

River pollution usually takes place because of human activities. Sources of river pollution can be accidental or intentional. They include industry and agriculture, such as the use of pesticides and fertilisers and the release of sewage into waterways.

Both river flooding and pollution can have social, economic, and environmental impacts (Table 1.2). These are usually small in scale, affecting local or regional areas. They can also have larger-scale implications, such as the destruction of crops for export to other countries. Whilst the impacts of flooding and pollution are largely negative, flooding can be positive as sediment deposited by floodwater acts as a natural fertiliser for farmland and can be beneficial for migrating species of birds.

**Table 1.2:** Impacts of river flooding and pollution

| | **Flooding** | **Pollution** |
|---|---|---|
| Social | • risk to human health, such as exposure to sewage leaks and contaminated floodwaters<br>• displacement of people<br>• fatalities and injuries. | • threats to human health<br>• shortage of drinking water<br>• shortage of water for irrigation. |
| Economic | • damage to property and disruption to infrastructure, leading to economic losses<br>• damage to crops, leading to economic losses. | • cost of cleaning up polluted watercourses and restoring affected ecosystems. |
| Environmental | • habitat destruction and soil erosion caused by the force of the floodwater. | • damage to aquatic ecosystems and loss of biodiversity. |


**Figure 1.8:** River contaminated by mine effluent and raw sewage, Bolivia

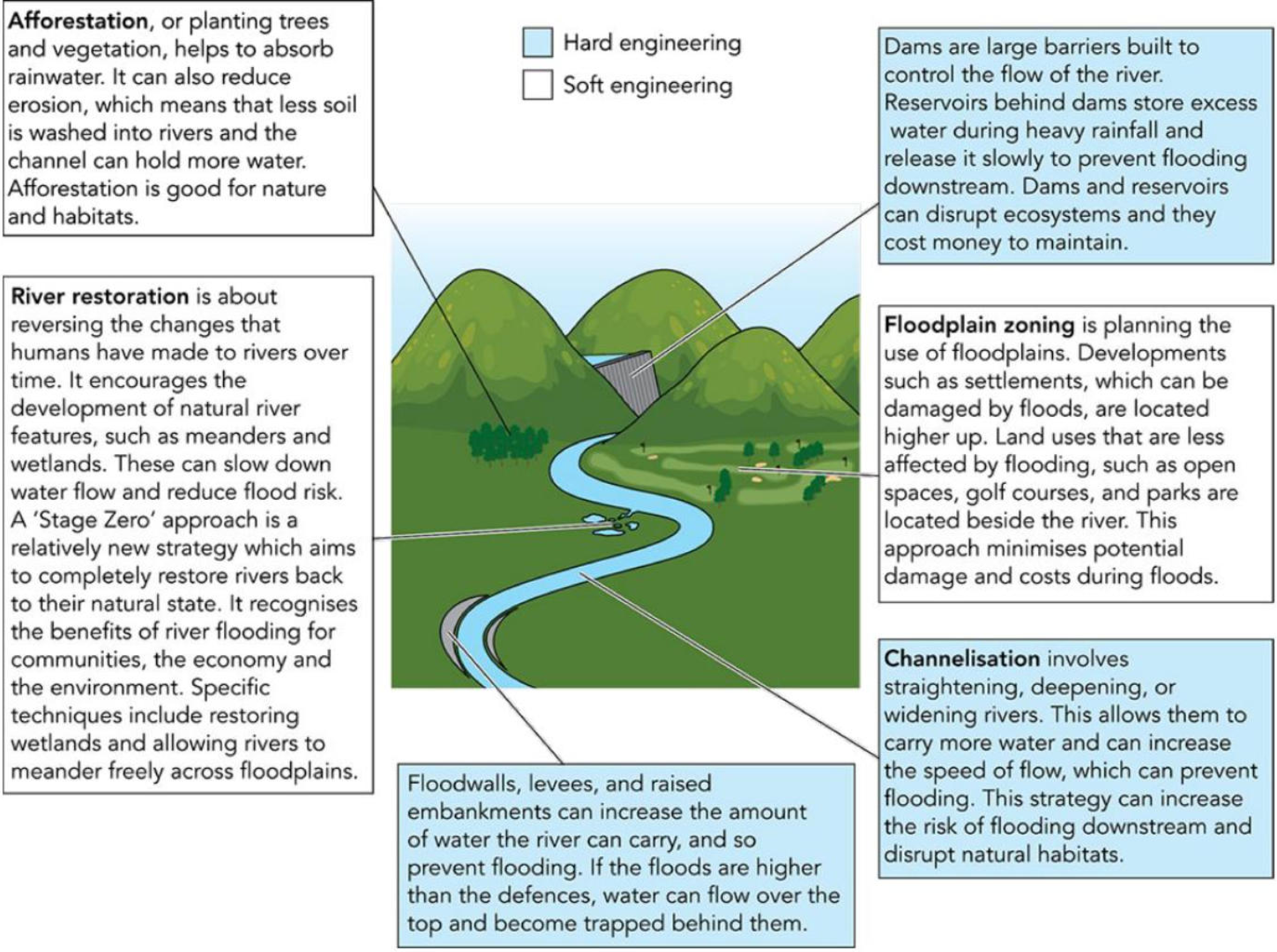
## Sustainable river management

Humans can implement strategies (management plans and agreements) and techniques (management methods) to reduce the risk of flooding and to ensure water quality in rivers. Sustainable river management involves both strategic planning and the use of specific techniques, whilst also considering environmental conservation, such as restoring and protecting the natural landscape and ecosystem.

Flood prevention strategies involve agreements to zone floodplains to prevent development or plans for river restoration. Techniques to prevent flooding include **hard engineering** and **soft engineering**. Hard engineering strategies tend to reduce flooding very quickly, but they can be expensive to build. Soft engineering strategies are generally considered to be more sustainable as they cause less disruption to nature.

**TIP**

It can be easy to confuse hard and soft engineering techniques. Hard techniques often use physically hard materials, such as concrete and steel. Soft techniques generally use softer materials, like vegetation. This rule doesn't always work – for example floodplain zoning does not use 'materials' at all – but it can be a good place to start.




**Source C**: Strategies and techniques to reduce flooding

Managing river pollution ensures water quality and protects ecosystems. Strategies for managing river pollution include the implementation of laws to protect rivers. This protection can be achieved using several different techniques:

- treating wastewater, such as sewage, before releasing it into rivers
- monitoring the quality of water in rivers to detect and deal with pollution
- fining companies and individuals that release contaminated water
- educating companies and individuals about the dangers of water pollution.

**Figure 1.9:** Water treatment facility, USA

## TASK 4: Study Source C

In groups, carry out a class debate. Half of the class should be 'for' hard engineering and the other half should be 'for' soft engineering strategies. You must consider social, economic, and environmental impacts of the techniques. Within your groups, you could take on specific roles (for example, residents, local business owners, tourists, environmentalists, the local council, and engineering companies).

## REFLECTION

How does Source C help you to remember and understand hard and soft engineering strategies and techniques? What other information from the 'river flooding and pollution' section could be turned into a diagram?

## THINK LIKE A GEOGRAPHER

Many people work in jobs planning and building flood defences. What could their roles be? These jobs can be within local communities or at a government level, and in some places, they are becoming more important because of the impacts of climate change. What types of skills might be needed for this line of work? Do you have those skills?